Escape speed

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain escape in an ideal gravitational-energy model.

SUCCESS CRITERIA - I can...

1. State the zero-at-infinity energy convention.
2. Distinguish escape speed from circular-orbit speed.
3. Name neglected effects in the ideal calculation.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

Total kinetic energy changes from 15 J to 12 J in an isolated collision. Is it elastic? Where could the difference go?

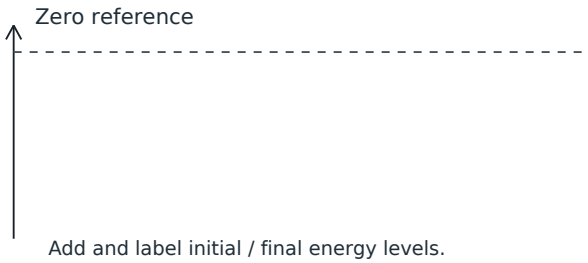
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $v_{escape} = \sqrt{(2\ GM/r)}$; $v_{circular} = \sqrt{(GM/r)}$. Ideal spherical body; ignore atmosphere/rotation. Minimum escape reaches infinity at zero speed with no further thrust.

Worked reasoning: For a simulated planet with $GM = 50\ m^3/s^2$ and radius 4 m, escape speed is $\sqrt{(2\ GM/R)} = 5$ m/s. At that limiting speed the ideal final speed at infinity is zero.

Your turn: Draw initial kinetic and gravitational-potential-energy bars whose total is zero.



Add and label initial / final energy levels.

Label K/m, U/m and total E/m in J/kg.

Marzano check - circle: ① ② ③

03 Read and interpret the model

Does escaping mean that gravity turns off at the launch point? Explain what the speed allows in an energy model.

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

04 Apply the physics

A model needs 50 J to reach infinity. A projectile begins with 60 J kinetic energy. What energy is left at infinity, neglecting losses? Use the energy model to show the quantities you compare; label units and any unknowns.

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

Escape speed

antocciphysics

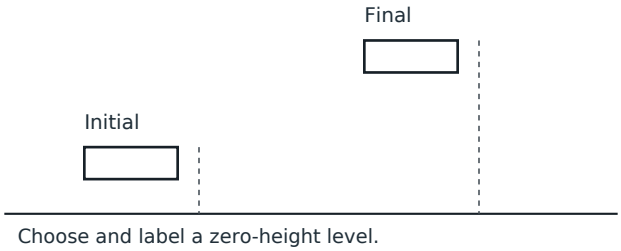
05 Explain your reasoning

Why is an ideal escape-speed calculation not a complete rocket-launch plan?

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Lift a 6 kg object by 8 m. Use g=10 N/kg. Find the change in gravitational potential energy.
Label the distances or energy levels used in your reasoning.

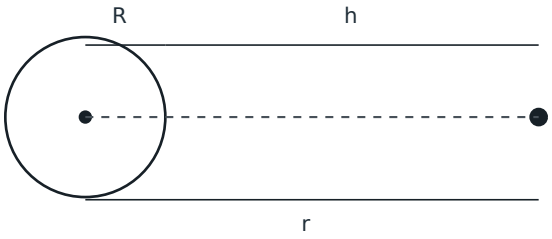


Choose and label a zero-height level.

Marzano check - circle: 1 2 3

07 Exit check

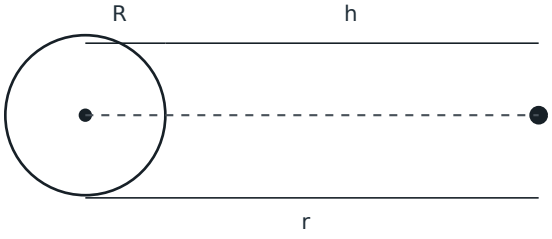
An object has less energy than required to escape and receives no further thrust. Can this ideal model expect it to escape? Use the energy model to show the quantities you compare; label units and any unknowns.



Marzano check - circle: 1 2 3

08 STRETCH: test the model

The same simulated planet's circular orbital speed at its surface is supplied as 3.54 m/s. Compare it with escape speed 5 m/s and explain their different purposes. Use the energy model to show the quantities you compare; label units and any unknowns.



Marzano check - circle: 1 2 3